Yipeng (Harry) Wang

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Academic Interests

I am an instructor at Princeton University. My research interests lie in geometric analysis and various aspects of differential geometry. Currently, I am focused on investigating problems related to minimal surfaces and the geometry of manifolds with positive scalar curvature.

Current and Forthcoming Positions

- Instructor, Department of Mathematics, Princeton University (2026–2027).
- Member, School of Mathematics, Institute for Advanced Study (2027–2029).

Education

Columbia University

- Ph.D. in Mathematics (2022–2026).
- Advisor: Simon Brendle.

Columbia University

- M.A. in Mathematics, 2024
- B.S. in Applied Mathematics, *Magna Cum Laude* (Class of 2022).

Publications and Preprints

1. **On a Variant of the Penrose Conjecture** (joint with Sven Hirsch). arXiv:2604.26046
2. **On the spacetime positive energy theorem in arbitrary dimension** (joint with Simon Brendle). arXiv:2604.18561
3. **A dimension descent scheme for the positive mass theorem in arbitrary dimension** (joint with Simon Brendle). arXiv:2604.08473
4. **Uniqueness of Cylindrical Tangent Cones $\mathbf{C_{p,q}} \times \mathbf{R}$** (joint with Benjy Firester, Raphael Tsiamis). Calc. Var. 65, 165 (2026). DOI: 10.1007/s00526-025-03238-5
5. **Area-minimizing capillary cones** (joint with Benjy Firester, Raphael Tsiamis). arXiv:2601.18794
6. **Stability inequalities for one-phase cones** (joint with Benjy Firester, Raphael Tsiamis). arXiv:2601.16966
7. **A sharp estimate for fill-ins with scalar curvature bounded from below** (joint with Simon Brendle, Raphael Tsiamis). arXiv:2510.17780
8. **Rigidity Results Involving Stabilized Scalar Curvature**. arXiv:2501.06951

[1] Updated August 24, 2026

9. **On Gromov's rigidity theorem for polytopes with acute angles** (joint with Simon Brendle). J. reine angew. Math. 2025. DOI: 10.1515/crelle-2025-0048
10. **Fill-Ins of Tori with Scalar Curvature Bounded from Below**. arXiv:2411.14667
11. **Effective volume growth of three-manifolds with positive scalar curvature**. Proc. Amer. Math. Soc. 154 (2026), no. 1, 329–337. DOI: 10.1090/proc/17082
12. **Scalar curvature rigidity of warped product metrics** (joint with Christian Bär, Bernard Hanke and Simon Brendle). SIGMA 20 (2024), 035. DOI: 10.3842/SIGMA.2024.035

Expository

1. **Scalar Curvature Rigidity of Polytopes**. Oberwolfach Report 9/2024

Invited Talks

1. *The Positive Mass Theorem in Arbitrary Dimensions.* Geometric Analysis and Topology Seminar, Courant Institute, New York University, September 2026.
2. *On the spacetime positive energy theorem in arbitrary dimension.* Differential Geometry and Geometric Analysis Seminar, Princeton University, September 2026.
3. *The Positive Mass Theorem in Arbitrary Dimensions.* Differential Geometry and Geometric Analysis Seminar, Princeton University, September 2026.
4. *The Positive Mass Theorem in Arbitrary Dimensions.* Geometry Seminars and Colloquia, Lehigh University, September 2026.
5. *The Positive Mass Theorem in Arbitrary Dimensions.* ICM Satellite Conference on Conformal Geometry, Einstein Manifolds, and Minimal Submanifolds, Penn State University, July 2026.
6. *The Positive Mass Theorem in Arbitrary Dimensions.* CUNY Geometric Analysis Seminar, CUNY Graduate Center, May 14, 2026.
7. *The Positive Mass Theorem in Arbitrary Dimensions.* Peking–Westlake Geometric Analysis Seminar, Online, May 14, 2026.
8. *The Positive Mass Theorem in Arbitrary Dimensions.* 2025–2026 Geometric Analysis Colloquium, Fields Institute, Toronto, May 2026.
9. *The Positive Mass Theorem in Arbitrary Dimensions.* Geometry & Topology Seminar, McMaster University, May 2026.
10. *The Positive Mass Theorem in Arbitrary Dimensions.* Geometry/Topology Seminar, Rutgers University, April 2026.
11. *The Positive Mass Theorem in Arbitrary Dimensions.* Geometry/Topology Seminar, Stony Brook University, April 2026.
12. *Fill-in Estimate with Scalar Curvature Bounded from Below.* Geometric Analysis, BICMR, Peking University (online), October 2025.
13. *Fill-In Problems with Scalar Curvature Constraints.* Convergence of Scalar Curvature Workshop, Online, July 2025.
14. *Fill-In Problems with Scalar Curvature Constraints.* Oberseminar Differential Geometry, Universität Augsburg, May 2025.
15. *Rigidity Results Involving Stabilized Scalar Curvature.* Geometric Analysis Seminar, CUNY Graduate Center, May 2025.

16. *Rigidity Results Involving Stabilized Scalar Curvature.* Geometric Analysis, Westlake University (online), April 2025.
17. *Fill-In Problems with Scalar Curvature Constraints.* Geometry Seminar, Stanford University, January 2025.
18. *Rigidity Results in Scalar Curvature.* Geometric Analysis and Topology Seminar, Courant Institute, New York University, November 2024.
19. *Rigidity Results in Scalar Curvature.* Seminar on Pure Mathematics, Hong Kong University of Science and Technology, June 2024.
20. *Rigidity Results in Scalar Curvature.* CHANGE — CHallenges in ANalysis and GEometry, ETH Zürich, June 2024.
21. *Scalar Curvature Rigidity of Polytopes.* Analysis, Geometry and Topology of Positive Scalar Curvature, Mathematisches Forschungsinstitut Oberwolfach, February 2024.
22. *On Gromov's rigidity theorem for polytopes with acute angles.* Not Only Scalar Curvature Seminar (GNOSC), Online, October 2023.

Teaching Experience

Princeton University

- Instructor, Calculus (Fall 2026)

Columbia University

- Teaching Assistant, Modern Analysis I (Spring 2026)
- Teaching Assistant, Fourier Analysis (Fall 2025)
- Instructor, Calculus II (Summer 2025)
- Teaching Assistant, Ordinary Differential Equations (Spring 2025).
- Teaching Assistant, Ordinary Differential Equations (Fall 2024).
- Teaching Assistant, Partial Differential Equations (Spring 2024).
- Teaching Assistant, Fourier Analysis (Fall 2023).
- Teaching Assistant, Calculus IV (Spring 2022).
- Teaching Assistant, Fourier Analysis (Fall 2021).

Organization

Seminars

- Co-organizer of *Student Geometric Analysis Seminar: Lagrangian Mean Curvature Flow*, with Raphael Tsiamis (Fall 2025). Seminar website
- Co-organizer of *Student Geometric Analysis Seminar: Min-max theory for minimal submanifolds*, with Raphael Tsiamis (Spring 2025). Seminar website
- Co-organizer of *Student Geometric Analysis Seminar: Regularity Theory of the Monge-Ampère Equation*, with Raphael Tsiamis and Jingbo Wan (Spring 2024). Seminar website.

- Co-organizer of *Student Geometric Analysis Seminar: Mean Curvature Flow*, with Raphael Tsiamis and Jingbo Wan (Fall 2023). Seminar website.

- Co-organizer of *Student Geometric Analysis Seminar: Dihedral Rigidity*, with Aaron Chow and Jingbo Wan (Spring 2023). Seminar website.

Programming Skills

Mathematica, Python, C++